TONER-1

260,032 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a little girl named Lily. She loved to play outside in the park. One day, she saw a big, red ball. She wanted to play with it, but it was too high. She asked her mom, “Can I play with the ball?” Her mom said, “Yes, it’s time to go home.”

Lily was very excited to see the ball. She put the ball in the ball and put it in the ball. But the ball was too high. Lily was sad. She did not want to share her ball. She said, “I will help you find it.”

Lily was sad and didn’t k…

TOKEN STREAM

All 200 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.

HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
200	**3.6**	**54.9**	**42** (unused)	**0** greedy

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	stories260K (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought.

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.